Walmart

You are a Data Analyst for the Walmart Vision Center team focused on understanding customer preferences and product performance in eyewear....

Question 1: Using sales data from January 2024, what are the top 3 eyewear styles by total units sold and what is the total number of units sold for each style? This query helps identify the most popular styles among customers.

```sql
SELECT style, SUM(quantity_sold) AS total_units_sold
FROM fct_sales AS f
INNER JOIN dim_products AS d
USING (product_id)
WHERE DATE_TRUNC('month',sale_date)::DATE = '2024-01-01'
GROUP BY style
ORDER BY 2 DESC
LIMIT 3
```

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Question 2: For each eyewear style sold in January 2024, what is the customer rating for the sale event that achieved the highest number of units sold? If there is a tie for highest units sold, return the event with the highest customer satisfaction rating. This query helps us understand if our most active buyers are happy with their purchase.

```sql
WITH calculate AS (
  SELECT d.style, MAX(f.quantity_sold) AS max_sold
  FROM fct_sales AS f
  JOIN dim_products AS d USING (product_id)
  WHERE sale_date >= '2024-01-01' AND sale_date < '2024-02-01'
  GROUP BY d.style
),

combine AS (
  SELECT
    f.sale_id,
    d.style,
    f.quantity_sold,
    f.customer_satisfaction,
    c.max_sold
  FROM fct_sales AS f
  JOIN dim_products AS d USING (product_id)
  JOIN calculate AS c
    ON f.quantity_sold = c.max_sold AND d.style = c.style
  WHERE sale_date >= '2024-01-01' AND sale_date < '2024-02-01')

SELECT *
  FROM(
    SELECT *,
      ROW_NUMBER() OVER (PARTITION BY style ORDER BY customer_satisfaction DESC) AS rank_satisfaction
    FROM combine) AS subquery
WHERE rank_satisfaction = 1
```

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Question 3: For each unique combination of eyewear style and price point in January 2024, calculate the product performance score. Product performance score is defined as the total units sold multiplied by the average customer satisfaction score. Which combination has the highest product performance score?

Note: to derive the price point for each sale, you can divide the sale_amount by the quantity_sold. This query will inform strategic decisions by identifying the mix of style and price point that best drives both sales and customer satisfaction.

```
with total AS(
    SELECT d.style, price,
          quantity_sold,
          ROUND(sale_amount / quantity_sold) AS price_point,
          customer_satisfaction
  FROM fct_sales AS f
  INNER JOIN dim_products AS d
  USING(product_id)
  WHERE sale_date >= '2024-01-01' AND sale_date < '2024-02-01')

SELECT total.style,
        price_point,
        SUM(quantity_sold) AS total_unit_sold,
        AVG(customer_satisfaction) AS average_customer_satisfaction_score,
        SUM(quantity_sold) * AVG(customer_satisfaction) AS product_performance_score
FROM total
GROUP BY total.style, price_point
ORDER BY product_performance_score DESC
LIMIT 1
```